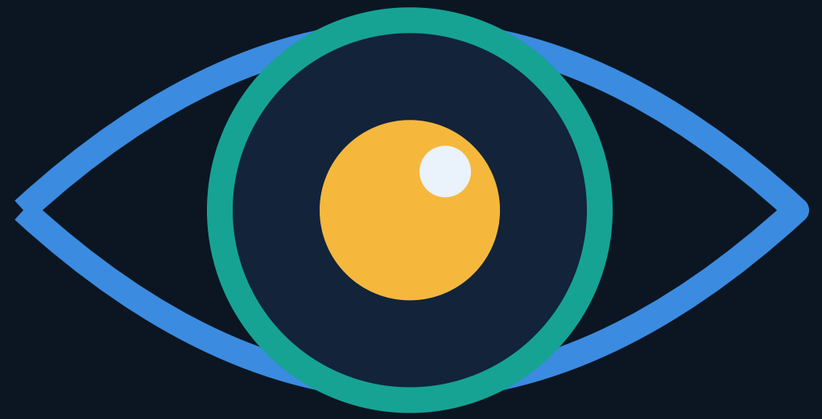
WEEK 4 · DAY 3

Give the Model Eyes 👁️

Feature engineering: new columns that let a model see what an engineer sees.



history • context • relationships

TODAY'S MISSION · 2 HOURS

Three kinds of eyes. Then your own method.



History

lags + rolling windows, no peeking ahead



Relationships

hot for its workload?



Fewer, better

32 columns vs 5



Checkpoint 3

your method, drafted

⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 55 min hands-on · 🏁 25 min checkpoint + wrap-up

2 MAY · STEADY SUNSHINE · EV02

NEPA brought light... but the inverter isn't charging 💡🔋

💡

The light is ON

Solar-array current: **26.93–27.14 A**.
Inside its normal sunlit band (26.71–29.26 A).
Looks fine.

🔋

The battery is stuck

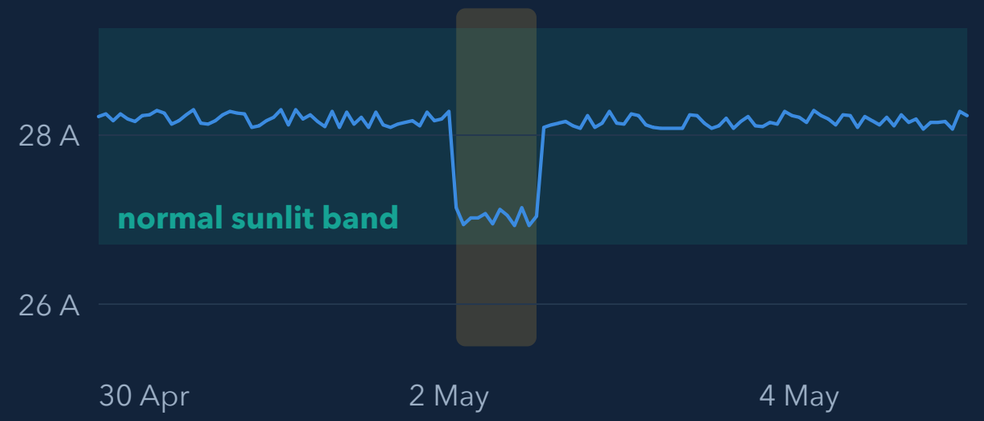
State of charge: **92.5–93.4%** for 12 hours.
On eclipse nights it reads 92–94% for 35 hours a year. **Normal... at night.**

Each number is normal **alone**. Together, in steady sun? **Only ever during a fault.** 🚨

30 APRIL → 4 MAY, HOUR BY HOUR

The current barely moves. The battery stops charging.

💡 Solar-array current



🔋 State of charge



gold = EV02, 2 May 01:00–12:00

bus voltage sags too: 50.12–50.24 V

the code book looked at 1 May by mistake (SoC 100% all day)

LAST TIME · THE FOREST ON RAW CHANNELS, STEADY SUN

4 of 5 caught… but look at EV04 👀

4/5
events “caught”

1/672
EV04 hours it noticed

25 days
late, the first (and only) time

For four weeks the amplifier ran hot **for its workload**.
The forest can't see “for its workload”. **Today we give it eyes.**

WHY FEATURES?

The model cannot tell time ⏰

```
13:00 · 27.0 A · 93.0%
14:00 · 27.1 A · 92.8%
15:00 · 27.0 A · 93.1%
```

rows of a table



judges ONE row at a time

No yesterday.
No “compared with what?”

🕰️

History

What was it an hour ago?
Over the last day?

🌒

Context

Steady sun, eclipse or recharge?

🔗

Relationship

Hot for its drive level?

PART 1

History, without cheating 🕰️

Give every row a memory of the past. Only the past.

FIVE LINES OF PANDAS

Every window ends now

```
soc.shift(1)                    # an hour ago
soc.shift(24)                   # same hour yesterday
soc.diff(1)                     # change since last hour
soc.rolling(24).mean()          # average of the last 24 hours
soc - soc.rolling(24).mean()    # how far from its own last day
```



THE ONE QUESTION TO ASK OF EVERY FEATURE

“Could I compute this at 14:00, using only what I knew at 14:00?”

✅ Honest

```
soc.shift(1)
soc.rolling(24).mean()
```

the past only

❌ Cheats

```
soc.shift(-1)
soc.rolling(24, center=True).mean()
```

reads tomorrow's newspaper today 🗞️

longest window 24 h **the first 24 rows have no full history** **8,760 → 8,736 rows**

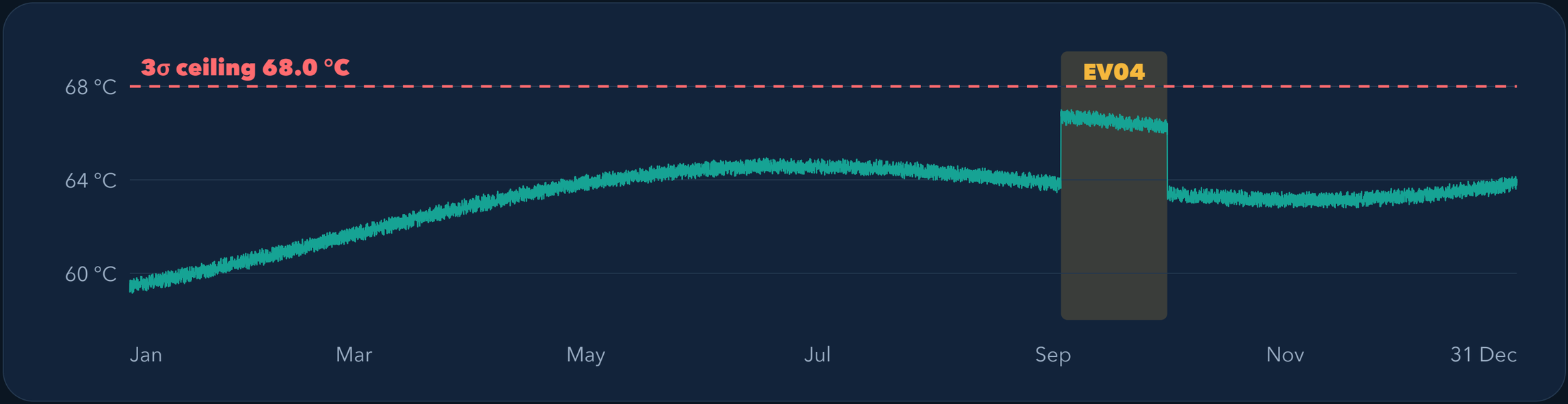
PART 2

Hot for its workload 🔗

One column that ties two sensors together.

EV04 · COOLING PATH DEGRADED · 3–30 SEPTEMBER

Raw temperature: four weeks, zero alarms



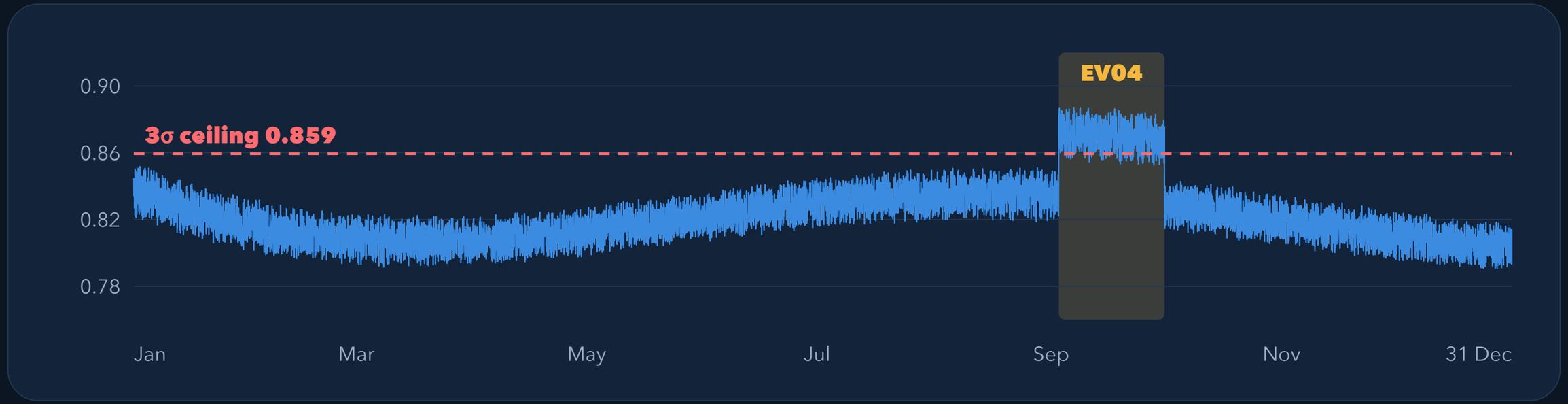
hottest EV04 hour 67.0 °C

ceiling 68.0 °C (Jan–Aug, steady sun)

the yearly swing hides a 3 °C fault

ONE NEW COLUMN: TEMP_PER_DRIVE = TWTA_TEMP_C ÷ RF_DRIVE_PCT

Divide by the workload and EV04 jumps out



normal 0.821 ± 0.013

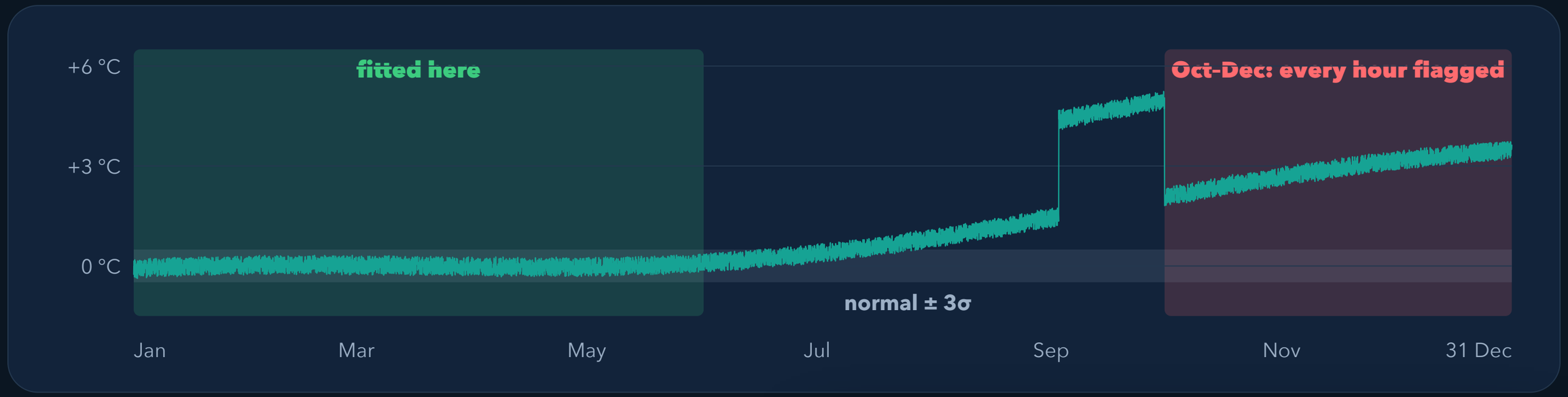
EV04 0.870 = +3.9σ

605 of 672 EV04 hours lit

0 false alarms all year

THE CODE BOOK'S FEATURE · TEMPERATURE VS DRIVE + SEASON, FITTED ON JANUARY-MAY

The 29σ trap



🧱 **Code book residual:** EV04 = **28.9σ**
but 2,208 of 2,208 Oct–Dec hours flagged · **3,512 false alarms**

➗ **Humble ratio:** EV04 = **3.9σ**
and **0 false alarms** all year

PART 3

More features ≠ better eyes ✂️

The code book's 32 columns against our 5.

THE CODE BOOK'S BUILDER · 32 COLUMNS FROM 6 SENSORS

18 pairs of near-twins 👯

Most alike (correlation)

`battery_soc_pct ↔ soc_gap` **1.000**

`solar_array_current_a_std24 ↔ bus_voltage_v_std24` **0.994**

`bus_voltage_v ↔ eclipse_flag` **0.994**

`twta_temp_c_lag1 ↔ twta_temp_c_mean24` **0.992**

`twta_temp_c ↔ twta_temp_c_mean24` **0.992**

soc_gap is just `100 − SoC`. Same column, wearing a different cap. 🧢

The forest picks a column **at random** for every cut. Two copies of SoC = SoC gets **double the vote**.

SAME FOREST · FITTED ON JAN-AUG STEADY SUN · SCORED ON THE WHOLE YEAR

32 columns vs 5 columns

EVENT	🪣 KITCHEN SINK (32)	🎯 LEAN 5 (4 SENSORS + TEMP_PER_DRIVE)
EV01 · point	❌ missed	✅ 1/1 h
EV02 · correlation break	✅ 11/12 h	✅ 12/12 h
EV03 · point	❌ missed	✅ 1/1 h
EV04 · thermal decoupling	✅ 410/672 h	✅ 358/672 h
EV05 · correlation break	✅ 24/24 h	✅ 24/24 h
False alarms · whole year	about 70	40

The sink misses the two **easiest** faults: an 11 A drop drowned in 32 columns of noise.

EXPLAIN EVERY FLAG · Z = (VALUE − NORMAL) ÷ Σ

Right answer, right reason

EVENT	WHAT BROKE	BIGGEST Z-SCORES OF THE FLAGGED HOURS
EV01	array string dropout	**array current** −33.8σ · **battery temp** +24.8σ
EV02	battery not charging	**state of charge** −21.6σ · **bus voltage** −5.9σ
EV03	array string dropout	**array current** −34.2σ · **battery temp** +27.5σ
EV04	amplifier hot for its drive	**temp_per_drive** +3.8σ
EV05	battery not charging (again)	**state of charge** −21.6σ · **bus voltage** −5.5σ

The model's reason matches the engineer's reason. That's the sentence you give the duty engineer. 👩🏾‍🔧

SO… IS IT READY FOR THE CONTROL ROOM?

5 of 5! 🎉 But hold on…



40 false alarms

about one page every 9 days



Marked its own homework

EV01–EV03 sit inside the months it trained on



No race yet

the Week 4 chart never got the same eyes

Next time 👉 **No expo!** An honest exam on months the model has never seen.

🏁 FINAL PROJECT · CHECKPOINT 3

🛠️ Method drafted

Bring today

✅ the minimum method for your brief, **running** on your data
✅ your "normal" in writing: states and what you take away first
✅ your features, each passing the 14:00 test
✅ your reference period: quiet months, agreed
✅ the simple baseline you'll race

Data rules

⛔ real NIGCOMSAT data stays on NIGCOMSAT systems (training server / work laptop), **not Colab**
⛔ no raw data on GitHub
⛔ describe the shape to AI tools, don't paste it

Next checkpoint: **Week 5 • Day 3: honest test** ⚖️

YOUR TURN · 55 MINUTES

Build the eyes 👁️

1. Open **W4D3-exercise**, save a copy, run STEP 0
2. **A** — History features + the 14:00 test
3. **B** — temp_per_drive, then spring the 29σ trap
4. **C** — 32 columns vs 5, and explain every flag



15 checks to pass

🌶️ Pepper task: give the old Week 4 chart the same eyes. Who wins?

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know



The model can't tell time. Lags and rolling windows give it memory. Past only: the 14:00 test.



One relationship beats many raw columns. Check every feature on months it never saw.



Fewer, sharper features. Prune twins; explain every flag with z-scores.

Next time 👉 **No expo!** Honest train/test splits + Quiz 3. 📝